

GameFlow Affordances: Towards a Tool for Designing Gameful Experiences

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ABSTRACT

Following publication in 2005, the original GameFlow model has seen thousands of citations and hundreds of applications to designing and evaluating games and gameful experiences. In the intervening years, the GameFlow model has been put to the test and validated by applying it to different game experiences to further understand those experiences and to expose any weaknesses of the model. This paper builds upon the history of applications, validation, and interrogation of the GameFlow model to redevelop and propose a new version of GameFlow, called *GameFlow Affordances*. GameFlow Affordances shifts GameFlow from a prescriptive list of low-level criteria, to a modernised, high-level model for affording enjoyable experiences. We review the previous commentary and recommendations on the GameFlow model, propose a new model in response to previous work, and outline next steps in developing a design tool for using GameFlow Affordances in the design of games and gameful experiences.

CCS CONCEPTS

• Human-centered computing → Heuristic evaluations; • Software and its engineering → Interactive games.

KEYWORDS

video games, enjoyment, player experience, GameFlow, heuristics, evaluation, design

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1 INTRODUCTION

GameFlow [42] is a model of player enjoyment, consisting of 38 criteria derived from games user experience literature and structured into eight elements that conceptually map to Csikszentmihalyi's [12] concept of flow. In the original 2005 GameFlow paper, Sweetser and Wyeth [42] identified the key elements of player enjoyment

in video games by conducting a comprehensive review of the literature on usability and user experience in games. The result was the identification of eight core elements of player enjoyment in games – Concentration, Challenge, Player Skills, Control, Clear Goals, Feedback, Immersion, and Social Interaction. These core elements were mapped to the elements of flow [12] after conceptual similarities were observed. This structure formed the foundation of a new model of player enjoyment in games, called GameFlow. Each element of the GameFlow model consists of an overall goal and a set of criteria that can be used to design and evaluate games with respect to player enjoyment. Since the original publication of the GameFlow model in 2005, it has been extensively used throughout games research and development communities, as well as more broadly for gameful design of non-game applications [33].

In 2019, Sweetser and Johnson [35] investigated the relationship between GameFlow and other player experience measures. They analysed the overlaps and divergences between the conceptual constructs of GameFlow and selected player experience measures and the different roles each instrument could play in game development. Sweetser and Johnson [35] observed that ratings alone are unlikely to provide a complete understanding of why and how a game is providing an enjoyable experience or to provide all available insights to a developer who is trying to improve their game. They proposed that, given the already extensive number of player experience measures available for researchers, a key opportunity for use and future development of the GameFlow model lies in supporting developers and researchers to inquire into how and why a game is, or is not, providing, or *affording*, an enjoyable experience, via heuristics. Sweetser and Johnson [35] concluded that GameFlow provides a unique contribution as a design and evaluation tool for in-depth inquiry, which works alongside validated player experience measures, and that valuable future work lies in further developing the GameFlow model to this purpose.

In 2020, Sweetser [33] aimed to identify where, why, and how GameFlow has been applied. They surveyed 2445 papers and identified 205 applications of GameFlow over 15 years (2005–2020) and reported on the types of applications, platforms, domains, audience, and usages of the model. Sweetser [33] found that half of the applications of GameFlow were to Serious Games, with most of these relating to education and training, followed by health and fitness. About a third of the applications were to games with the primary purpose of entertainment. The remaining applications were to non-game applications and experiences. The survey demonstrated the extensive usage of the GameFlow model beyond the model's original purposes of designing and evaluating games for enjoyment. Sweetser [33] concluded that future versions of the GameFlow

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Table 1: Extensions and derivations of the GameFlow model.

Model	Description
EGameFlow	A questionnaire developed to evaluate e-learning games [15]
Pervasive GameFlow (PGF)	A model to evaluate player enjoyment of pervasive games [20, 21]
RTS-GameFlow	A model developed specifically for real-time strategy games [14]
ARG-PGF	An extension the PGF model to apply to alternate reality games [26, 27]
Social GameFlow	A model to conceptualise how games might foster flow and cooperative learning through social play [2]
MIU-GameFlow	A model for enjoyable playing experiences for motor impaired users [44]
Others	A number of other models and measures draw on GameFlow along with other constructs, including the Playful Experience Framework [25] and the Instructional Game Evaluation Framework [43]

Table 2: Previous work to test the GameFlow model.

Context	Method
Online strategy game under development	Evaluation with different stakeholders (developer, players, journalists, researchers) by conducting expert reviews using Gameflow, alongside other player experience measures [34–36].
VR versus non-VR games	Evaluation by conducting expert reviews using GameFlow criteria and analysing game reviews using GameFlow as a theoretical framework [40, 41].
First person shooter games on console, adventure games on mobile	Evaluation by conducting expert reviews using GameFlow criteria [38].
Real-time strategy games	Analysing professional game reviews using GameFlow as a theoretical framework [37, 39].
Real-time strategy games on PC	Evaluation by conducting expert reviews using GameFlow criteria [42].

model should take into consideration the broad range of applications, with a particular focus on Serious Games for education and health.

Previous work has also evaluated and critiqued the GameFlow model [3, 10, 18, 19, 30] and tested the applicability of the model to different contexts to identify various weaknesses of the model [34–40]. In this paper, we report on our current research, which builds on the history of previous GameFlow research, applications, and validation to develop a revised model and tool that addresses the previous recommendations and identified issues, and to ensure that it is fit for purpose and fulfilling a proven need in research and development communities. We present a new model, called *GameFlow Affordances*, that shifts GameFlow from a set of elements and criteria, to a model of seven affordances for designing games and gameful experiences to afford player enjoyment and satisfaction. The new model incorporates the history of GameFlow learnings and represents a modernisation of the model, with the intention for GameFlow Affordances to form the basis of a new game design tool for game developers and researchers that can be integrated into a playcentric game design process [16].

2 FLOW AND GAMEFLOW

Csikszentmihalyi’s concept of flow [12] seeks to understand what makes experiences enjoyable and why people engage in certain activities. Csikszentmihalyi began his research with people who spend large amounts of time and effort on activities that are difficult, but provide no external rewards, such as composers, chess players, and rock climbers [11]. His later studies were conducted with a broader set of people, asking them to describe how it felt when their lives were at their fullest and when what they did was most enjoyable. He found that optimal experience, or flow, is the same around the world and irrespective of social class, age, or gender. Very different activities are described in similar ways when they are being enjoyed. Flow is an experience “so gratifying that people are willing to do it for its own sake, with little concern for what they will get out of it, even when it is difficult or dangerous” [12]. The key element in flow is that the experience is an end in itself – the activity is intrinsically rewarding and autotelic. In flow, the goal is often an “excuse” for the process [28].

Due to the lack of existing frameworks for evaluating enjoyment in games at the time (in the early 2000s), Sweetser and Wyeth [42] conducted a comprehensive review of the literature on usability and user experience in games to determine the key elements of player

Table 3: Previous recommendations for improvements to GameFlow [34, 36].

Type	Recommendation
Criteria format	For evaluation purposes, the criteria should be written as statements about the game against which the reviewer can rate their agreement, rather than heuristic-style statements of what games should provide.
Criteria wording	Unclear words should be removed or made clear in the context of the statement (i.e., “stimuli” in Concentration criteria and “errors” in Control criteria).
Criteria wording	Each criterion should be simplified to contain a single statement that can be given a rating, which will involve splitting some criteria (e.g., criteria in Concentration, Player Skills, Social Interaction).
Criteria wording	Criteria should be revised to remove language that is outdated or game-type-specific (e.g., “manual” in Player Skills criteria, “shell” and “starting, stopping, saving” in Control criteria).
Elements	The inclusion of the Immersion element needs to be reconsidered, with the likely removal of this element or, otherwise, significant reconceptualisation.
Process	It should be made clearer that elements and criteria are optional and that not all criteria apply to all games and that criteria can be given an N/A rating.
Criteria wording	The criteria should be unified in whether they are referring to the game, the player, the player experience, or what the game provides or affords.

Table 4: Predominant player experience measures.

Measure	Description
PXI (2020)	The Player Experience Inventory measures player experiences at the level of functional and psychosocial consequences [1]
GIQ (2015)	The Game Immersion Questionnaire includes engagement, engrossment, and immersion [8]
GEQ (2009)	The Game Engagement Questionnaire measures engagement in video game-playing based on absorption, flow, presence, and immersion [5]
IEQ (2008)	The Immersive Experience Questionnaire measures immersion in games based on involvement, dissociation, challenge, and control [22]
GEQ (2007)	The Game Experience Questionnaire measures user experience in games including competence, immersion, flow, tension, challenge, and affect [19]
PENS (2006)	The Player Experience of Need Satisfaction is based on self-determination theory [32]

enjoyment in video games. They identified eight core elements of player enjoyment in games: Concentration, Challenge, Player Skills, Control, Clear Goals, Feedback, Immersion, and Social Interaction. Sweetser and Wyeth [42] observed that these core elements overlapped closely with the elements of flow and subsequently mapped their core elements of player enjoyment to the elements of flow. This structure formed the foundation of a new model of player enjoyment in games, called GameFlow.

The GameFlow elements are all closely interrelated and interdependent. In summary, games must hold the player’s attention through a high workload, with tasks that are sufficiently challenging to be enjoyable. The player must be skilled enough to undertake the challenging tasks, the tasks must have clear goals so that the player can complete the tasks, and the player must receive feedback

on progress towards completing the tasks. If the player is sufficiently skilled and the tasks have clear goals and feedback, then the player will feel a sense of control over the task. The resulting feeling for the player is total immersion or absorption in the game, which causes the player to lose awareness of everyday life, concern for themselves, and alters their sense of time. The final element of player enjoyment, social interaction, does not map to the elements of flow, but features strongly in the player experience literature [4, 13]. Each element of the GameFlow model consists of an overall goal and a set of criteria that can be used to design and evaluate games with respect to player enjoyment.

The GameFlow model was designed to be a general model of player enjoyment, applicable to all game genres and platforms [38]. A number of extensions and derivations of the GameFlow model

have been developed to target GameFlow towards more specific experiences (see Table 1). In addition to the many applications of the GameFlow model, there has been a set of previous work that has aimed to test the applicability of the model to various contexts and to identify any weaknesses of the model (see Table 2). The GameFlow model has also been evaluated and critiqued in the literature, particularly focusing on the inclusion of the Social Interaction element [10], the relationship between Social Interaction and Immersion [18], the division between Player Skills and Challenge [3], the conceptualisation and measurement of Immersion [3, 19], and the GameFlow model's connection to the concept of flow [10, 30]. Previous research has found that the Immersion criteria are difficult to assess via expert review and that they might not apply to all types of games [39, 42]. It has also been identified that some GameFlow elements relate to the game and some relate to the player experience [38]. A number of recommendations have been made for improvements to GameFlow [34, 36] (see Table 3).

GameFlow was originally developed in recognition of the absence of an accepted model of enjoyment in games [42]. The goal of GameFlow was to integrate the existing disparate heuristics into a validated model that can be used to design, evaluate, and understand enjoyment in games. In the intervening years, there have also been a number of different models and measures that have emerged (see Table 4) for player experience. It is therefore important that any changes and updates to GameFlow take into consideration the full landscape of player experience tools currently available. In this paper, we build on the full history of GameFlow development, applications, and interrogation to propose an updated model that responds to key criticisms and aims to provide a modern tool for in-depth inquiry as part of the game development process.

3 GAMEFLOW AFFORDANCES

Based on the history of applications, validation, and interrogation of the GameFlow model and in response to the findings and critique, we propose the new model: *GameFlow Affordances* (see Table 5, Figure 1). GameFlow Affordances includes seven *affordances* (rather than elements), which reflects that game designers do not (and can not) directly create gameplay experiences. Rather, the experience exists between the player and the game [16]. Designers create the conditions for experiences to emerge when the player plays the game. The experience itself depends on many factors [23] (e.g., personality, mood, personal preferences, context of play), which are often outside of the designer's control. Sweetser et al. [38] identified that the GameFlow criteria are inconsistent in whether they are referring to the game, player, or experience. The new set of affordances all relate to the player experience, and how designers can design games to afford an enjoyable player experience.

Many of the issues identified in previous work [34, 36] related to specific criteria within GameFlow elements. These issues included the format of the criteria (heuristic-style statements, double-barrelled statements) or language of the criteria (unclear, outdated, game-specific). GameFlow Affordances does not include specific or prescriptive criteria within each affordance. Instead, we used thematic and conceptual analysis to extract the core sentiment across the sets of criteria to define the high-level affordances, with

regard to the critical analysis from the literature and key recommendations developed over previous works. Sweetser [33] found that many applications of GameFlow only used the high-level elements, rather than the specific criteria. We believe that the high-level affordances are the most useful component for game designers and researchers, particularly when seeking a general model that can be applied across different genres and platforms [39], and also while seeking to survive the test of time [34, 36]. The purpose of the affordances is to support designers in deep inquiry throughout the game design process, rather than providing a checklist or prescriptive set of guidelines or heuristics for design and/or evaluation. The affordances seek the heart of what it means to create an enjoyable or satisfying gameful experience. Our accompanying tool/process (via our ongoing work) will provide designers with the means to use the affordances for inquiry throughout an iterative playcentric game design process [16].

One of the key criticisms of the GameFlow model is that it includes the characteristic flow experience via the Immersion element [3, 10, 19, 30, 39, 42]. The Immersion element describes the 'deep but effortless involvement' that players experience while in the flow state when playing games, along with experiencing an 'altered sense of time' and loss of awareness of self and surroundings. Previous research has found that the flow experience is very difficult to measure or assess [24, 31] and it cannot be assessed via a third party to the experience (e.g., expert review) [39, 42]. In light of the criticisms and the refocusing of the model on how game designers can afford (design for) experiences, we removed Immersion from the model. At the same time, we refocused the original Concentration element to capture the balance between player limits (cognitive, physical, emotional) and game tasks and stimuli, via the Involvement affordance. Previous work [37, 39, 42] revealed that the Immersion element was being satisfied by various stimuli (e.g., sounds, graphics, narrative), and so this experience is now being channelled through the Involvement affordance, which captures both attention-demanding aspects (i.e., tasks) and sensory stimulating aspects (i.e., stimuli).

Another criticism of the GameFlow elements related to the division between Player Skills and Challenge [3]. The new model includes a Skill-Challenge Balance affordance (in place of the previous Challenge element), which maps to the central corresponding component of flow. This is still separate from the integration of learning and play, and the need for mastery and reward, via the new Learning & Mastery affordance. However, Challenge is not represented without connection to player skills. Most of the previous Player Skills criteria are captured within Learning & Mastery. However, pace of increasing players skills fits within Skill-Challenge Balance, while usability-focused criteria (e.g., of controls/interface) conceptually fit in the new Control & Agency affordance. Again, each of the new affordances are focused on how designers can afford enjoyable player experiences at a high-level, without being prescriptive about how to do so.

We revised the Control element from GameFlow to the Control & Agency affordance. Control relates to lower-level and direct control (e.g., over movement, interaction, interface, controls, play sessions), whereas agency relates to choice and meaningful impact in the game. We reconceptualised the Clear Goals and Feedback elements as the Goals & Progression affordance, with part of the Feedback

Table 5: Proposed new GameFlow model: GameFlow Affordances.

Affordance	Description
Involvement	The game affords player involvement by engaging the player (cognitively/physically/emotionally) with satisfying tasks/stimuli (quality, quantity, variety).
Skill-Challenge Balance	The game affords skill-challenge balance (for different players and throughout the game). Skill-challenge progression is bi-directional (increasing player skill drives challenge escalation and increasing challenge drives player skill development).
Learning & Mastery	The game affords player learning and mastery with seamless integration of learning and play throughout the game. The player is given opportunities to demonstrate their skill development and mastery and is rewarded appropriately.
Control & Agency	The game affords player control (over movement, interaction, interface, controls, play sessions) and agency (choice of actions/strategies, impact on game world). The player receives feedback on the success of their actions and strategies.
Goals & Progression	The game affords player goal setting and progression. Goals (overarching, intermediate) are clearly presented to the player and/or the player can set their own goals. The player receives feedback on progress towards goals and current status in the game.
Comfort & Safety	The game affords player comfort and safety (play space, body position, movements, wellness) while playing (for different players and throughout their play session).
Social Interaction	The game affords social interaction through social play (competition, cooperation, and/or observation) and social connection (inside and/or outside the game).

element being integrated into the Control & Agency affordance. This acknowledges that feedback is a device to provide visibility to players in how they are progressing towards their goals and to understand the impacts of their actions and strategies, thereby affording a sense of control and agency in the game.

The Social Interaction element has been maintained, as the Social Interaction affordance. There has been some previous criticism of the inclusion of social interaction in the model [10, 18]. However, the importance of social interaction to game (and human) experience has been widely acknowledged [4, 13]. Social Interaction is an interesting affordance, as it brings external influences (other people) into the game experience, adding a new dimension. In a similar vein, we have added the Comfort & Safety affordance, based on previous work related to VR games [40]. Like Social Interaction, Comfort & Safety relates to the player's connection to an external influence, their environment. It is applicable to many types of modern games, where the play space or location plays a key role in the experience, such as virtual/augmented/mixed reality, mobile, exercise, and location-based games.

The resulting seven GameFlow Affordances are: Involvement, Skill-Challenge Balance, Learning & Mastery, Control & Agency, Goals & Progression, Comfort & Safety, and Social Interaction (see Table 5). These affordances replace the eight elements in the 2005 GameFlow model: Concentration, Challenge, Player Skills, Control, Clear Goals, Feedback, Social Interaction, and Immersion. The 2005 GameFlow elements map to the GameFlow Affordances as shown in Figure 1. Each of the new affordances are written in the format 'The game affords ...' with a high-level statement of how each element of player experience can be supported. Additionally, each affordance

has a relationship with the game, player, interface, and external influences (i.e., other people or environment) (see Figure 1).

4 TOWARDS A GAMEFLOW DESIGN TOOL

GameFlow was originally conceived and presented as a set of heuristics for designing and evaluating games. The original GameFlow paper used an expert review method to validate the heuristics, proposing it as a qualitative tool for evaluation. The expert review entailed simple ratings against the criteria (0-5), along with analysis of how each criteria was met (or not) by the game. This method was repeated in subsequent papers [34–36, 38, 41]. Fifteen years later, Sweetser [33] found that researchers had mostly used it as a questionnaire, with others using it as an analytical tool, heuristics, or to structure interviews. However, as we discussed earlier, there are many other existing validated measures that can be used for quantitative questionnaires [1, 5, 8, 19, 22, 32]. The format in which GameFlow was presented (a list of 38 criteria structured into 8 elements) suggested that it should be used as a checklist or a survey. Sweetser [33] also found that researchers often reworded the criteria in order to use it as a survey, as the original criteria are phrased as what games/players 'should' do (heuristic-style), which researchers reworded into statements about the game under evaluation.

GameFlow Affordances is not a set of heuristics, it is a high-level conceptual model that aims to capture the essence of designing for enjoyable gameful experiences. We do not plan for it to be used as a checklist or survey or a usability tool to detect issues in completed games [29]. As such, we do not plan to release a list of accompanying heuristics or criteria for use in this way. Instead, we have identified

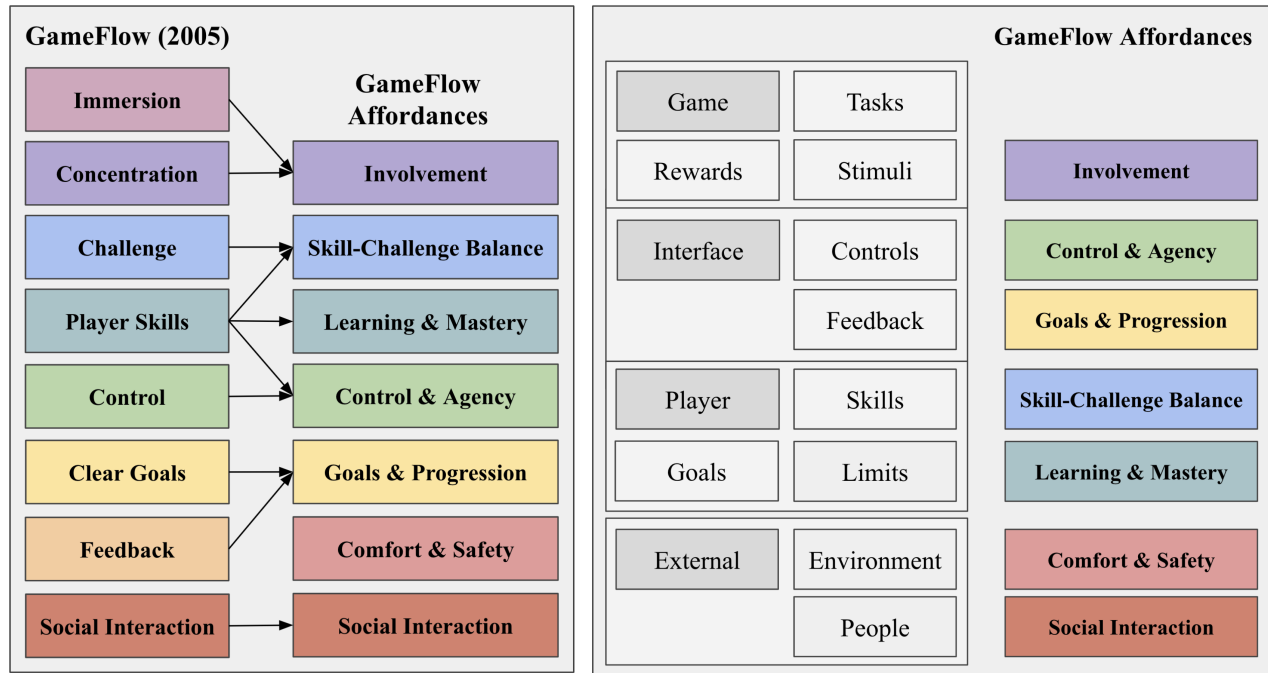


Figure 1: Mapping of original GameFlow model to new GameFlow Affordances (left) and new GameFlow Affordances model (right). Interface represents the conceptual overlap between the player and the game (rather than the game’s physical/digital interface). No arrows are drawn on the model, as the experience (affordances) is highly interconnected with the game/player/interface/external components (dark grey) and their sub-components (light grey), but proximity illustrates *approximate* conceptual closeness.

that GameFlow Affordances has the potential to be used as a design tool, by researchers and developers, in designing games and gameful experiences. Enjoyment in video games is a key driver of play [6, 7] and a predictor of game success [17]. Providing developers with a tool for designing games for enjoyment holds considerable value within the global video game market (expected to be worth US \$533 billion by 2027 [9]). Additionally, as identified by Sweetser [33], many researchers have applied GameFlow to experiences beyond games for entertainment. Our ongoing work aims to develop the tool (i.e., method by which GameFlow Affordances can be used) and to put it to the test in designing games and experiences to afford enjoyment. Ozdowska et al. [29] identified the importance of including the method in which games heuristics should be used with the release of new heuristics. The issues related to not providing a clear method are evidenced by the diverse applications of the original GameFlow model. Our design tool will be designed to complement and fit into an iterative playcentric game design process [16], including designing and evaluating player experience goals, game concepts, core gameplay, and formal elements. We will co-develop our method with game developers, with early versions tested in a university class with game developers in training, before being made available to professional game developers and researchers. We aim to support designers and researchers in understanding how to create gameful experiences, rather than providing another tool for post-hoc evaluation.

5 CONCLUSIONS

This paper presented a new incarnation of the GameFlow model, called *GameFlow Affordances*. We remodelled GameFlow based on the history of applications, validation, and interrogations published since 2005. GameFlow Affordances includes seven affordances, in place of the original eight GameFlow elements. The affordances embody high-level guidelines on designing enjoyable and satisfying games and gameful experiences. GameFlow Affordances represents a synthesis and modernisation of the original GameFlow elements and criteria, shifting away from low-level specific criteria. Our ongoing research uses GameFlow Affordances as the foundation for a game design tool that can be embedded in a playcentric design process. This tool will support game developers and researchers in designing enjoyable and satisfying games and gameful experiences for entertainment and beyond.

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